

KUDYMKAR, Russian FederationWMO: **281160**Lat: **59.017N**Lon: **54.650E**Elev: **186**StdP: **99.11**Time Zone: **5.00 (E05)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-32.0	-28.6	-34.9	0.1	-31.9	-31.3	0.2	-28.5	7.5	-10.6	6.7	-10.9	1.6	320	0.619

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.2	28.7	20.2	26.8	19.2	24.9	18.2	21.2	26.9	20.2	25.3	19.1	23.6	2.5	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.3	14.4	24.2	18.3	13.5	22.9	17.4	12.7	21.7	62.5	26.8	58.8	25.3	55.2	23.7	26.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7.4	6.5	5.7	-35.8	31.2	4.0	1.6	-38.7	32.4	-41.1	33.3	-43.4	34.2	-46.3	35.4
			-36.0	22.7	3.8	1.6	-38.7	23.9	-40.9	24.8	-43.0	25.7	-45.7	26.9
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	2.1	-13.9	-12.4	-4.8	3.0	10.0	15.1	17.5	14.9	9.2	2.4	-6.0	-11.4
	DBStd	12.39	7.84	7.53	4.90	5.09	5.43	4.32	4.24	4.27	4.46	4.64	6.63	7.84
	HDD10.0	3574	741	628	458	216	71	2	6	67	238	479	662	662
	HDD18.3	6021	999	862	716	459	262	114	67	124	274	493	729	921
	CDD10.0	678	0	0	0	6	71	162	236	157	43	4	0	0
	CDD18.3	84	0	0	0	0	4	18	43	18	1	0	0	0
	CDH23.3	760	0	0	0	1	59	154	391	149	7	0	0	0
	CDH26.7	176	0	0	0	0	9	26	98	42	1	0	0	0
	WSAvg	2.6	2.4	2.5	2.8	2.7	3.0	2.5	2.2	2.2	2.4	2.8	2.6	2.5
(15) Precipitation	PrecAvg	596	40	25	36	32	46	75	73	67	56	54	50	42
	PrecMax	762	66	59	77	63	101	144	160	151	126	135	110	69
	PrecMin	473	13	4	4	0	21	11	8	19	11	14	10	20
	PrecStd	76	14	12	19	18	21	35	40	31	27	25	21	13
(20) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	0.5	1.2	8.4	20.8	27.8	29.3	30.9	30.8	23.8	17.2	8.3	2.4
		MCWB	-0.1	0.4	4.0	11.8	16.5	20.5	21.2	21.2	15.8	11.1	7.2	2.0
	2%	DB	-0.6	0.4	4.9	16.4	24.9	26.9	29.2	27.1	20.4	13.5	4.9	1.0
		MCWB	-1.1	-0.2	1.3	10.3	15.1	19.3	20.6	19.8	15.0	9.7	4.1	0.4
	5%	DB	-1.9	-0.7	2.9	13.2	22.3	25.1	27.5	24.3	17.9	10.9	2.5	-0.1
		MCWB	-2.3	-1.3	0.5	7.8	13.8	18.3	20.1	18.5	14.0	8.6	1.9	-0.5
	10%	DB	-3.6	-2.3	1.5	10.6	19.7	22.9	25.5	22.1	15.9	8.9	1.1	-1.7
		MCWB	-4.0	-2.9	0.0	6.1	12.7	17.1	19.2	17.4	13.1	7.6	0.6	-2.2
	0.4%	WB	0.2	0.8	4.5	12.8	18.2	21.7	22.9	22.4	17.1	12.2	7.3	2.1
		MCDB	0.5	1.0	8.1	18.6	25.3	27.0	29.0	29.6	20.9	15.3	8.1	2.3
(28) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	-1.0	-0.1	2.1	10.6	16.3	20.4	21.7	20.4	15.8	10.2	4.2	0.5
		MCDB	-0.6	0.2	3.6	16.1	23.0	25.6	27.3	25.5	19.1	12.8	4.7	0.9
	5%	WB	-2.3	-1.4	1.1	8.4	14.7	19.2	20.7	19.1	14.5	8.7	2.1	-0.5
		MCDB	-1.9	-0.7	2.4	12.5	20.4	23.7	26.0	23.2	17.3	10.4	2.5	0.0
	10%	WB	-4.0	-2.8	0.1	6.6	13.3	17.8	19.7	17.9	13.2	7.5	0.7	-2.2
		MCDB	-3.6	-2.2	1.3	10.1	18.8	21.9	24.5	21.5	15.6	9.0	1.0	-1.7
	5% DB	MDBR	6.2	7.3	7.7	8.9	10.9	9.7	10.2	8.8	7.4	4.9	4.5	5.4
		MCDBR	5.2	5.1	8.5	12.9	14.8	12.3	13.0	12.1	10.3	8.5	4.4	3.8
		MCWBR	5.3	4.8	6.0	7.8	7.3	6.1	5.7	5.6	5.9	5.3	4.2	3.6
		MCWBR	5.5	5.0	6.7	12.1	13.2	11.0	11.7	11.2	9.2	7.7	4.3	3.6
(40) Clear-Sky Solar Irradiance	taub		0.278	0.309	0.358	0.385	0.379	0.373	0.399	0.403	0.363	0.334	0.293	0.265
		taud	2.004	2.026	1.969	2.104	2.337	2.397	2.332	2.301	2.381	2.305	2.107	1.964
	Ebn at Noon		487	673	747	809	850	861	825	789	764	661	473	358
		Edh at Noon	54	88	126	132	112	108	113	108	85	67	48	39
(44) All-Sky Solar Radiation	RadAvg		0.45	1.27	2.52	3.82	4.95	5.36	5.38	3.97	2.28	0.99	0.38	0.24
	RadStd		0.06	0.25	0.33	0.51	0.56	0.49	0.52	0.38	0.27	0.11	0.07	0.04

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (8 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page